



Sequential Logic - Flip Flop

Logic Design of Digital Systems (300-1209) section 1

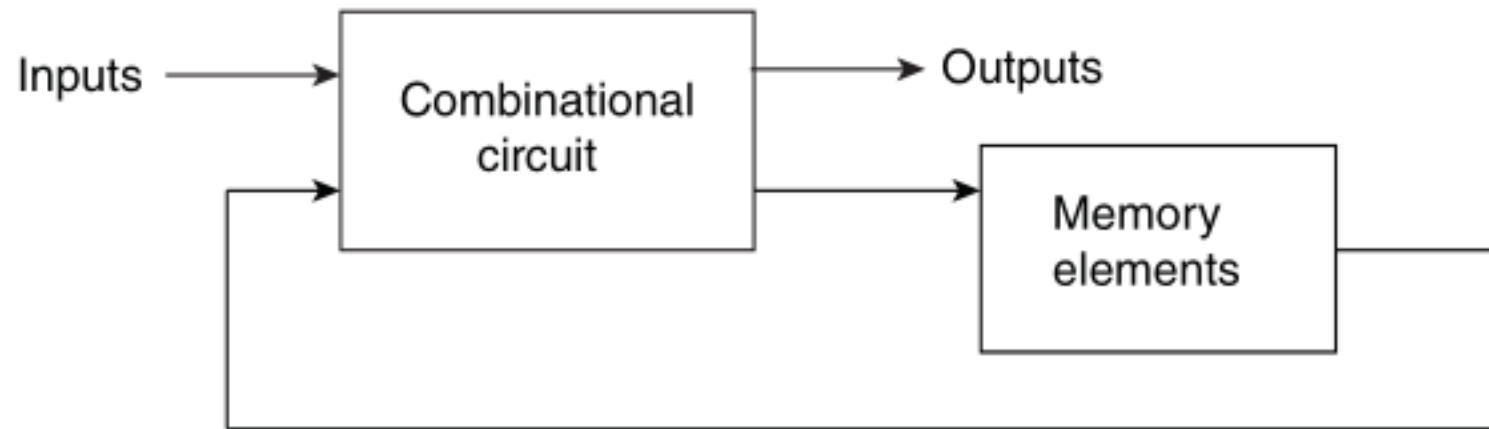
LECTURE 09

KRISADA PHROMSUTHIRAK

Sequential Logic Circuit

Combinational Logic: the outputs at any instant of time are entirely dependent upon the inputs present at that time.

Sequential Logic: the binary information stored in memory elements defines as a state of a sequential circuit. So, the **output not only depends on the input** but also the state of the circuit. Furthermore, the state may also change due to the input information.



Flip-Flop

It is a circuit that allows the digital world to **store binary information**.

For example:

if we count the numbers. We won't be able to count numbers if we don't know **how far we have counted**. In Programming, once we can count the numbers, we can do a loop program.

What flip-flops do is to maintain a binary state indefinitely until directed by an input signal to switch states.

One flip-flop can store 1 bit.

Each flip-flop has two outputs Q and Q' (Complement for each other)



Flip-Flop

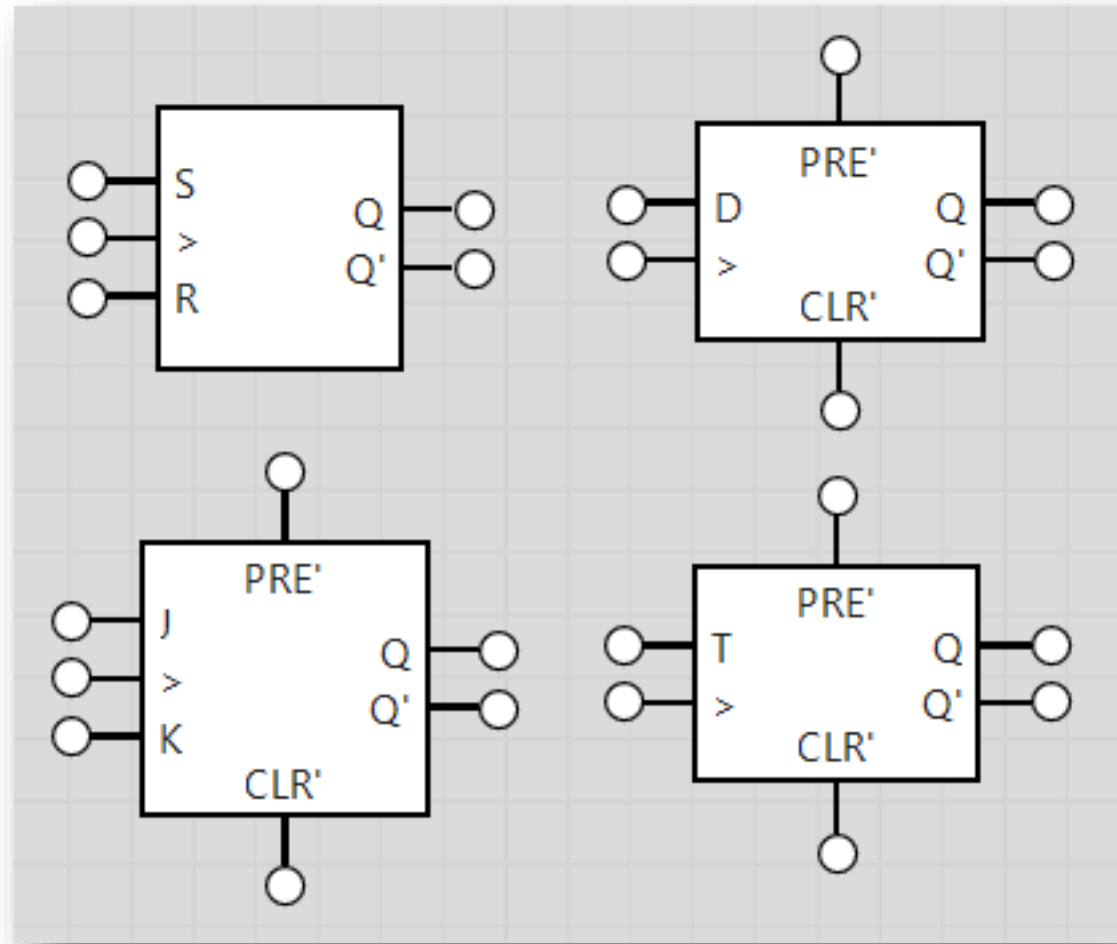
The input signal that we input to a flip-flop circuit can be either 1 Input, 2 Input, or more. It depends on the type of Flip-Flop.

Entering the input will cause the flip-flop to change its state, so we have two states that should be known: the present state and the next state.

Present state (คือสถานะ <u>ก่อน</u> ป้อนสัญญาณ Input)	$Q(t)$
Next state (คือสถานะ <u>หลัง</u> ป้อนสัญญาณ Input)	$Q(t+1)$

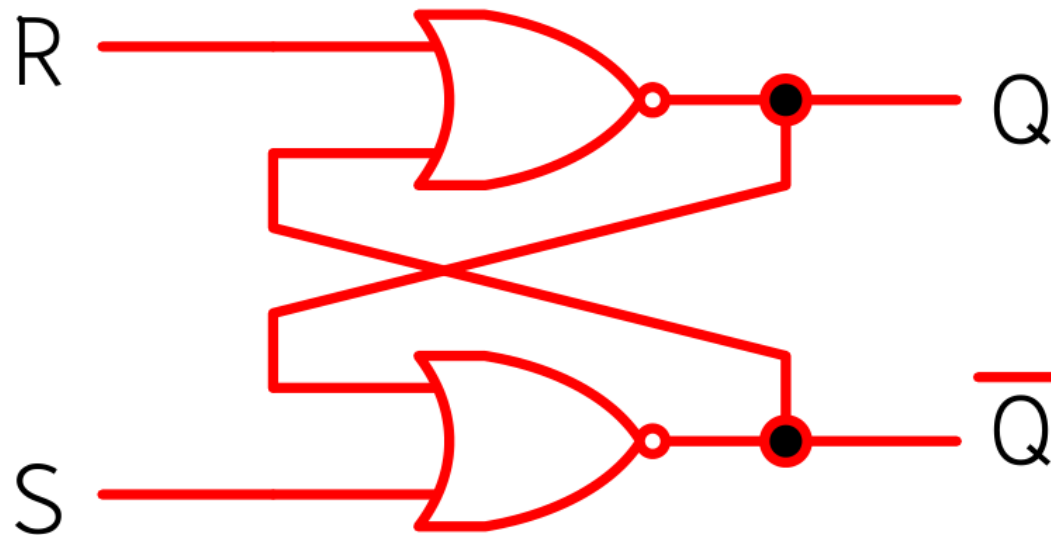
Type of Flip-Flop

- ☐ RS Flip-Flop
- ☐ JK Flip-Flop
- ☐ D Flip-Flop
- ☐ T Flip-Flop

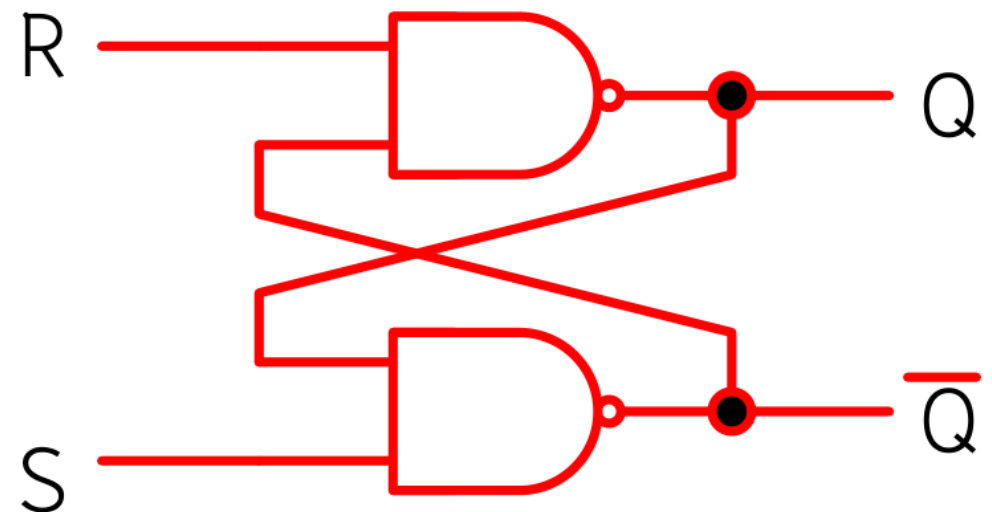


RS Flip-Flop

There are **two methods** for creating RS flip-flops using either a **NOR Gate** or a **NAND Gate**, which provide **different behaviors**.



NOR Gate RS Flip-Flop

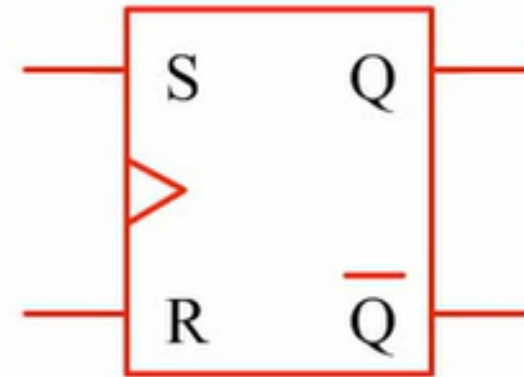
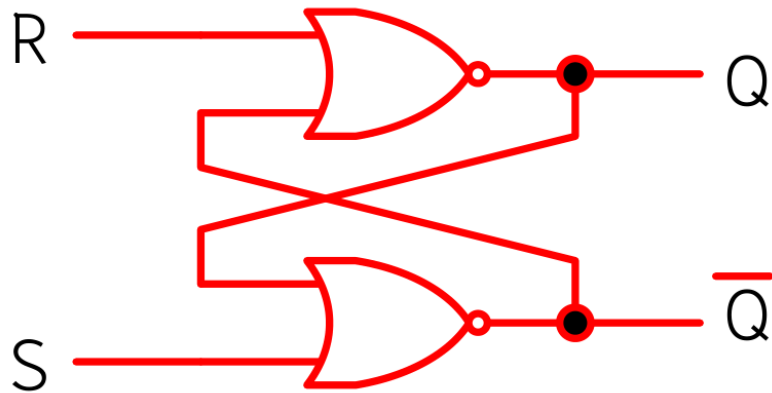


NAND Gate RS Flip-Flop

NOR Gate RS Flip-Flop

This RS Flip-Flop scheme uses two **NOR gates** to be crossed by the output to feedback to the input.

- ❑ Two input pins: **S (Set)** and **R (Reset)** used to control the state of the Flip-Flop.
- ❑ Two output pins: **Q and Q'** define a state of circuit that are complementary each other.



NOR Gate RS Flip-Flop

Program Simulation Online

<https://logic.ly/demo>

NOR Gate RS Flip-Flop

Characteristic

- ☐ Q กับ Q' เป็น Complement ของกันและกัน จะมีสถานะเดียวกันไม่ได้
- ☐ กำหนด S (Set) เป็น 1 จะทำให้ Output Q เป็น 1 Q' เป็น 0
- ☐ กำหนด R (Reset) เป็น 1 จะทำให้ Output Q เป็น 0 Q' เป็น 1
- ☐ กรณี S = 1 และ R = 1 วงจร ไม่สามารถทำงานได้
- ☐ กำหนดให้ Q(t) คือ Present state และ Q(t+1) คือ Next state
- ☐ NOR Gate RS Flip-Flop จะทำงานตาม Truth Table ต่อไปนี้

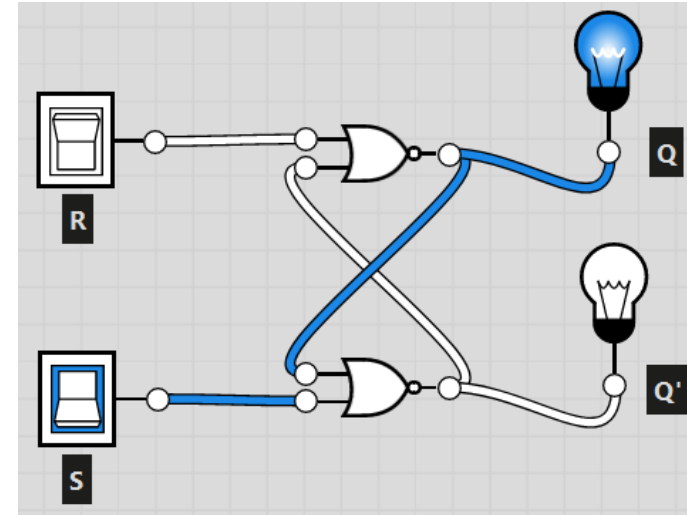
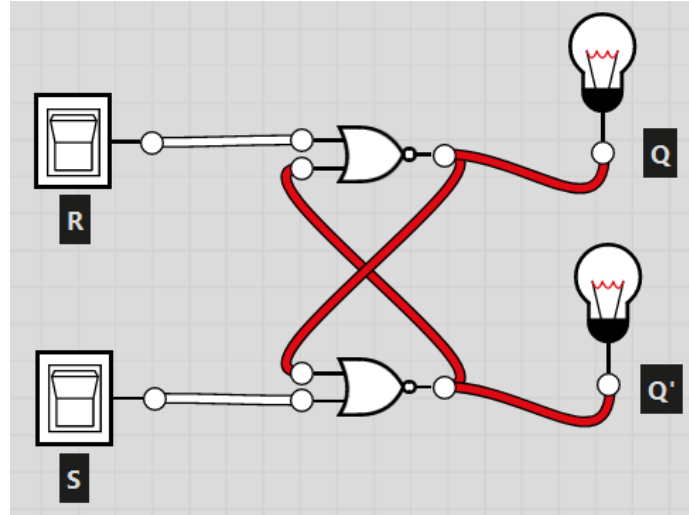
NOR Gate RS Flip-Flop

Characteristic

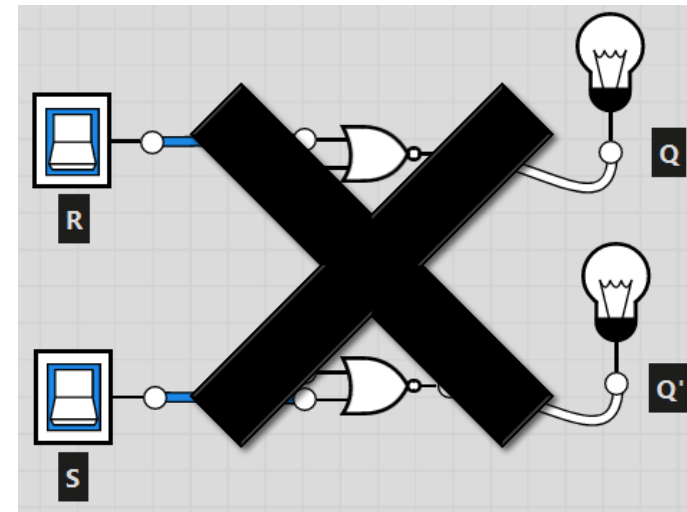
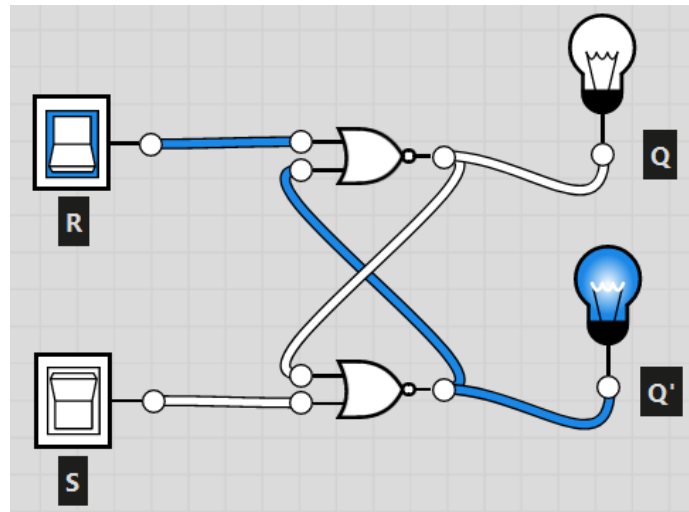
$S = 0$

$S = 1$

$R = 0$



$R = 1$



NOR Gate RS Flip-Flop

R	S	Q (t)	Q (t+1)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Characteristic Table



Characteristic Equation

NOR Gate RS Flip-Flop

Q (t)	Q (t+1)	R	S
0	0		
0	1		
1	0		
1	1		

Excitation Table
(ตารางการกระตุ้น)

R	S	Q (t+1)	Q' (t+1)
0	0		
0	1		
1	0		
1	1		

Operation Table
(ตารางการทำงาน)

NAND Gate RS Flip-Flop Characteristic

- ☐ Q กับ Q' เป็น Complement ของกันและกัน จะมีสถานะเดียวกันไม่ได้
- ☐ กำหนด S (Set) เป็น 1 จะทำให้ Output Q เป็น 1 Q' เป็น 0
- ☐ กำหนด R (Reset) เป็น 1 จะทำให้ Output Q เป็น 0 Q' เป็น 1
- ☐ กรณี S = 0 และ R = 0 วงจร ไม่สามารถทำงานได้
- ☐ กำหนดให้ Q(t) คือ Present state และ Q(t+1) คือ Next state
- ☐ NAND Gate RS Flip-Flop จะทำงานตาม Truth Table ต่อไปนี้

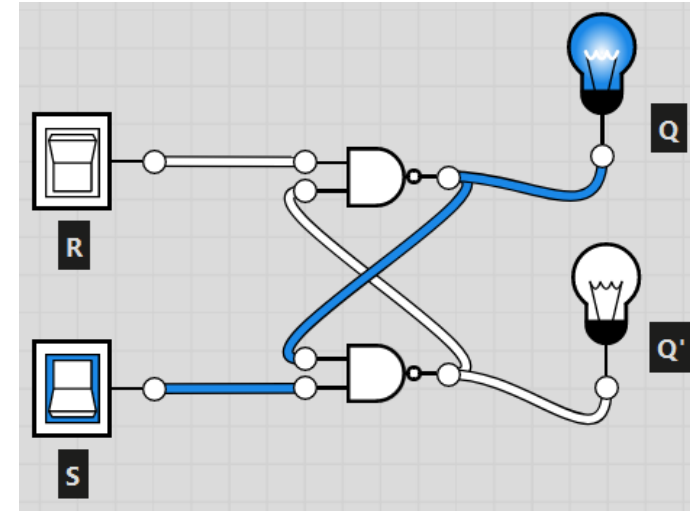
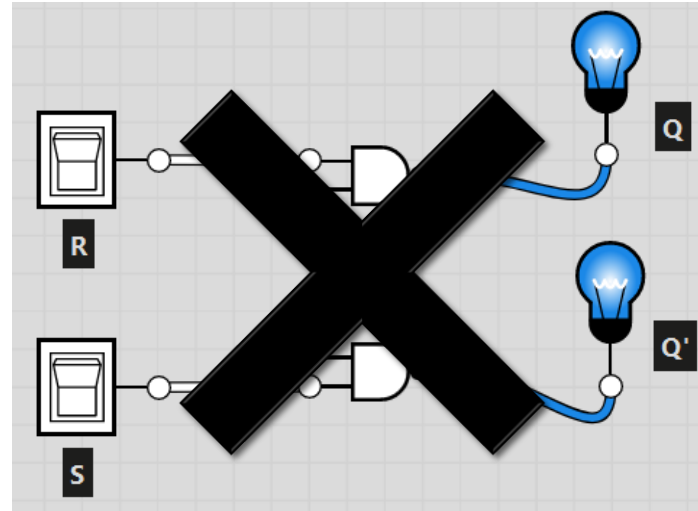
NAND Gate RS Flip-Flop

Characteristic

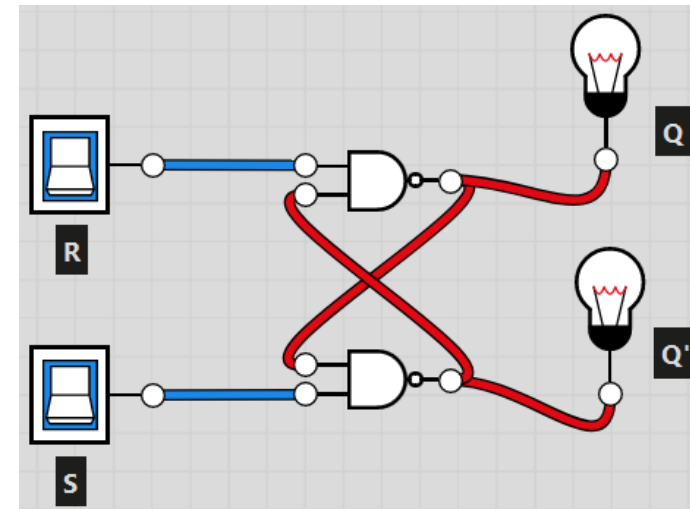
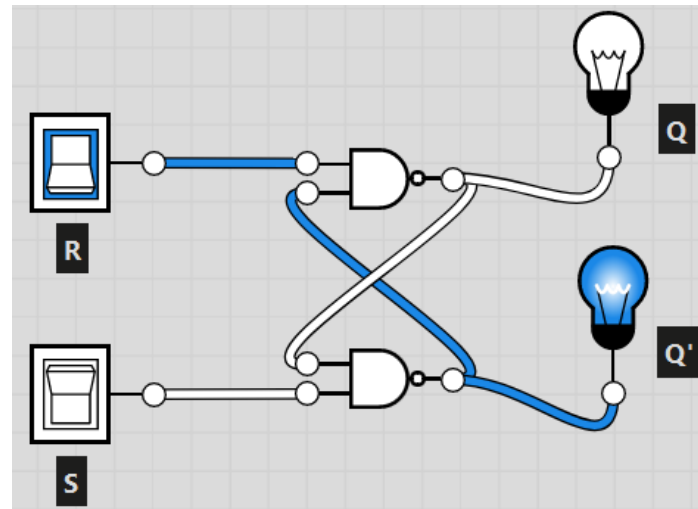
$S = 0$

$S = 1$

$R = 0$



$R = 1$



NAND Gate RS Flip-Flop

R	S	Q (t)	Q (t+1)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Characteristic Table



Characteristic Equation

NAND Gate RS Flip-Flop

Q (t)	Q (t+1)	R	S
0	0		
0	1		
1	0		
1	1		

Excitation Table
(ตารางการกระตุ้น)

R	S	Q (t+1)	Q' (t+1)
0	0		
0	1		
1	0		
1	1		

Operation Table
(ตารางการทำงาน)

Take a Break

REST YOUR MIND / RELAX

Clock RS Flip-Flop

The functionality of **Clock RS Flip-Flop** is similar to Program **RS Flip-Flop**, but with the difference that it adds **clock signal** to control time of state changing (Trigger).

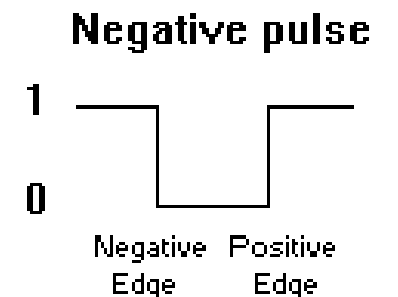
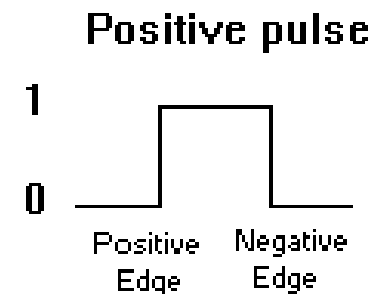
- ☐ Change state when there is a clock signal to stimulate circuit
- ☐ If no clock signal, maintain the current state.

Why we need clock signal ?

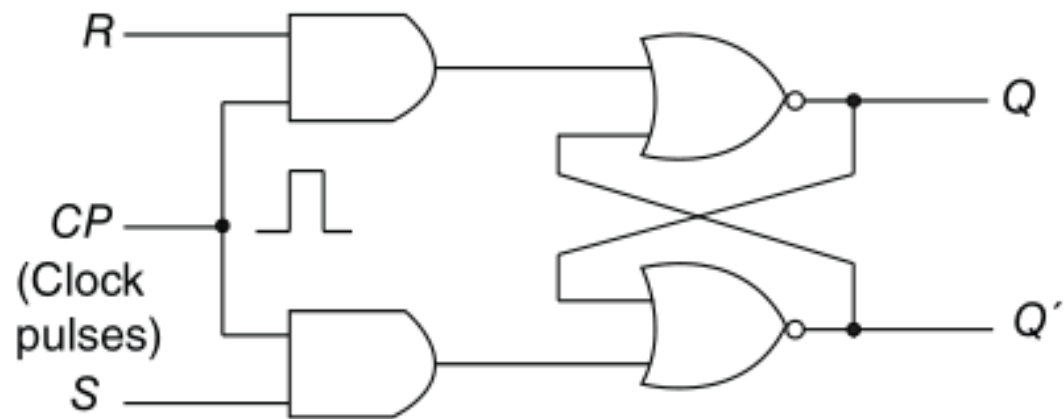
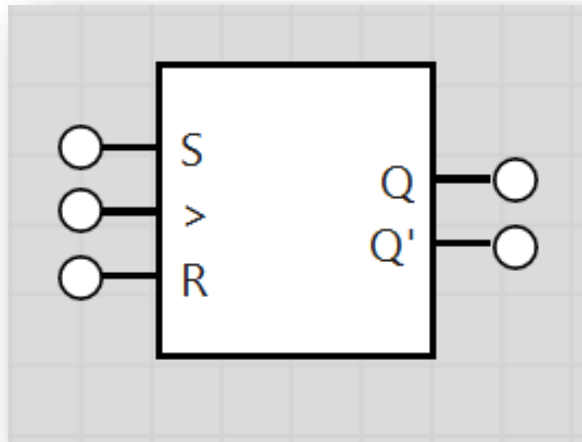
Clock Pulse (CP) or Clock Signal (CLK)

There are 4 ways to use clock pulses in a circuit.

- ☐ การกระตุ้นโดยใช้ขอบสัญญาณบวก (Positive-Edge Triggered)
- ☐ การกระตุ้นโดยใช้ขอบสัญญาณลบ (Negative-Edge Triggered)
- ☐ การกระตุ้นโดยใช้พัลส์บวก (Positive-Pulse Triggered)
- ☐ การกระตุ้นโดยใช้พัลส์ลบ (Negative-Pulse Triggered)



NOR Clock RS Flip-Flop

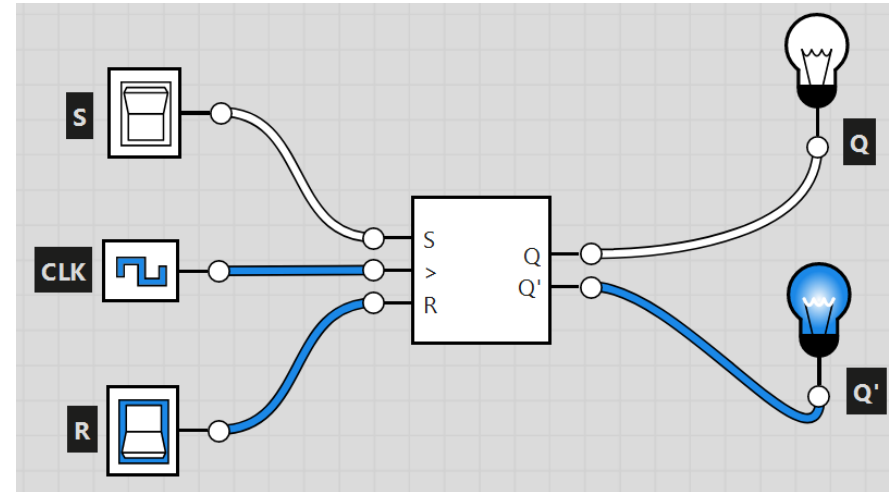
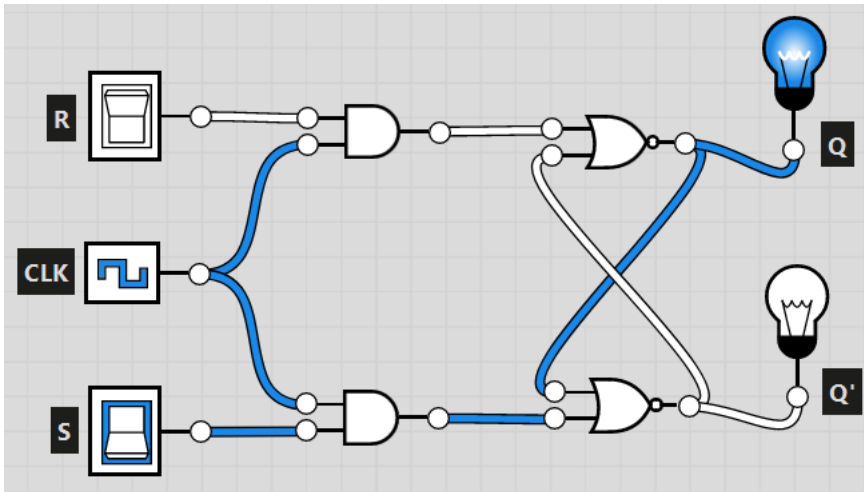
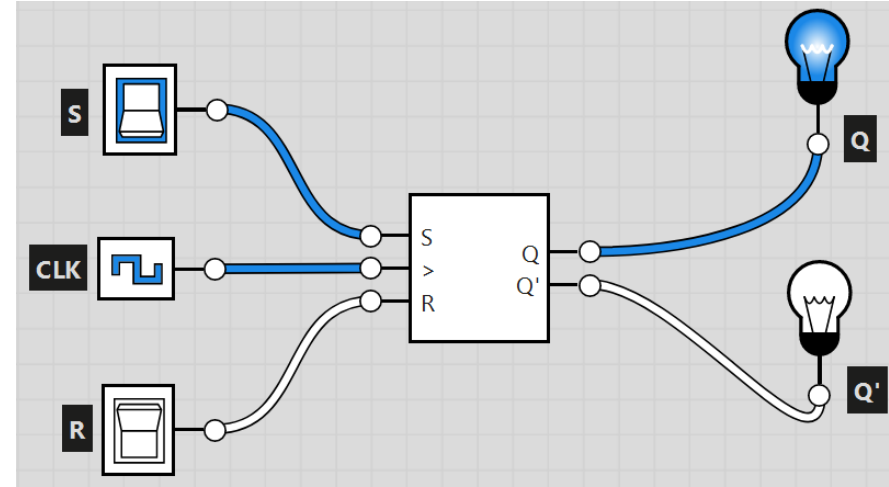
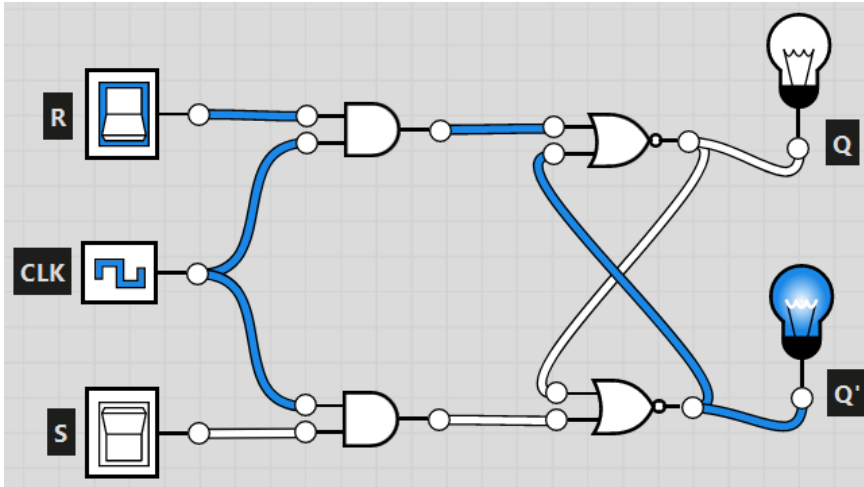


CP	R	S	Q (t+1)	Q' (t+1)
-	X	X		
↑	0	0		
↑	0	1		
↑	1	0		
↑	1	1		

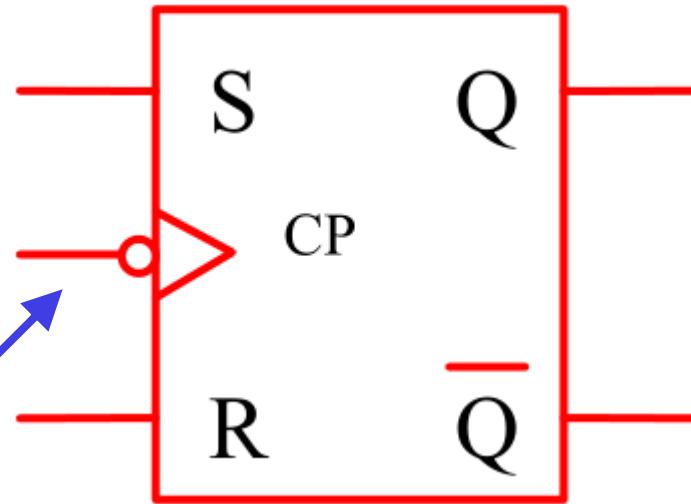
Operation Table
(ตารางการทำงาน)

NOR Clock RS Flip-Flop

ระวัง!!! ขาของ Input สลับกันนะ (แบบเขียนวงจร กับ แบบใช้ Symbol)

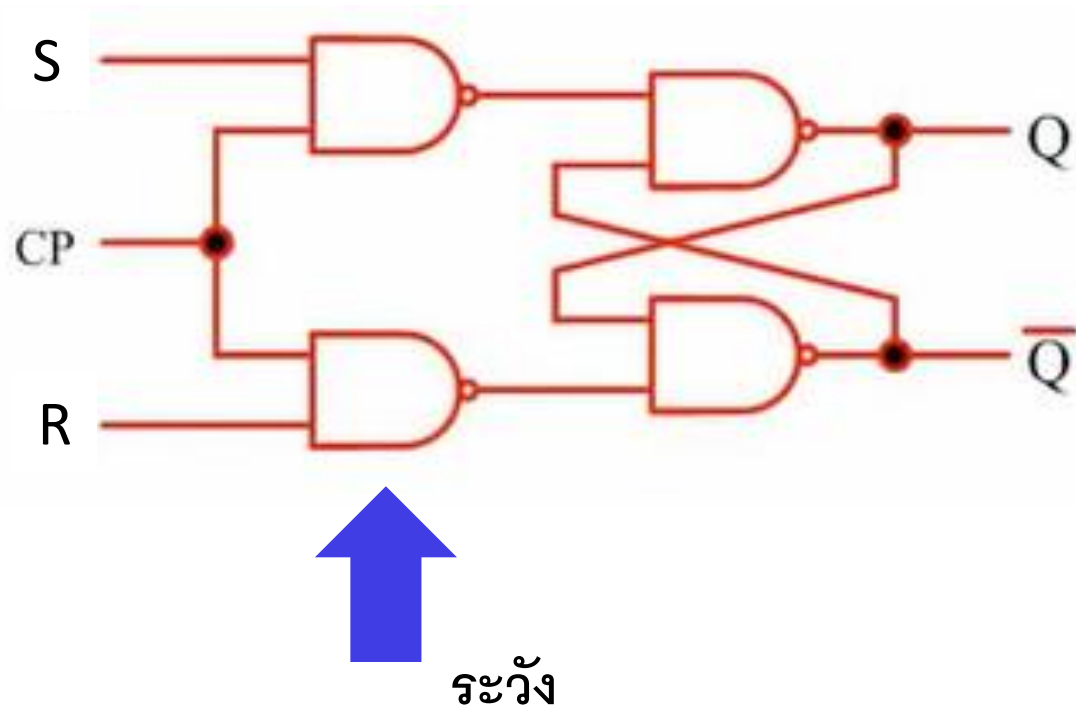


How to setup Clock Trigger with Negative



Negative-Edge Triggered

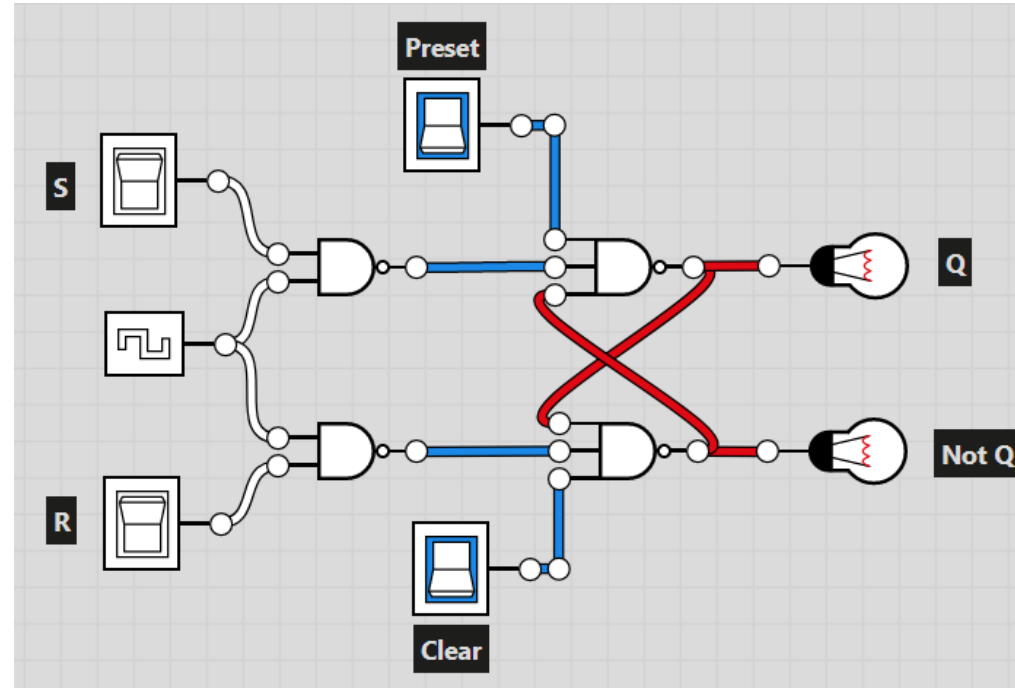
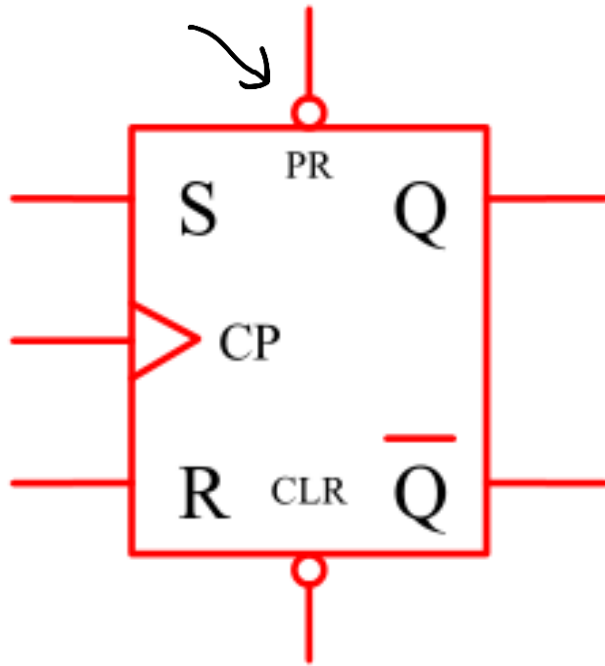
NAND Clock RS Flip-Flop



CP	R	S	Q (t+1)	Q' (t+1)
-	X	X		
↑	0	0		
↑	0	1		
↑	1	0		
↑	1	1		

Operation Table
(ตารางการทำงาน)

NAND Clock RS Flip-Flop with Preset (PR) & Clear (CLR)



[Preset] & [Clear] have higher priority than other input

ใส่ Clear เป็น 0 (ทำงานเมื่อใส่ 0) เข้าไป $\rightarrow Q = 0$

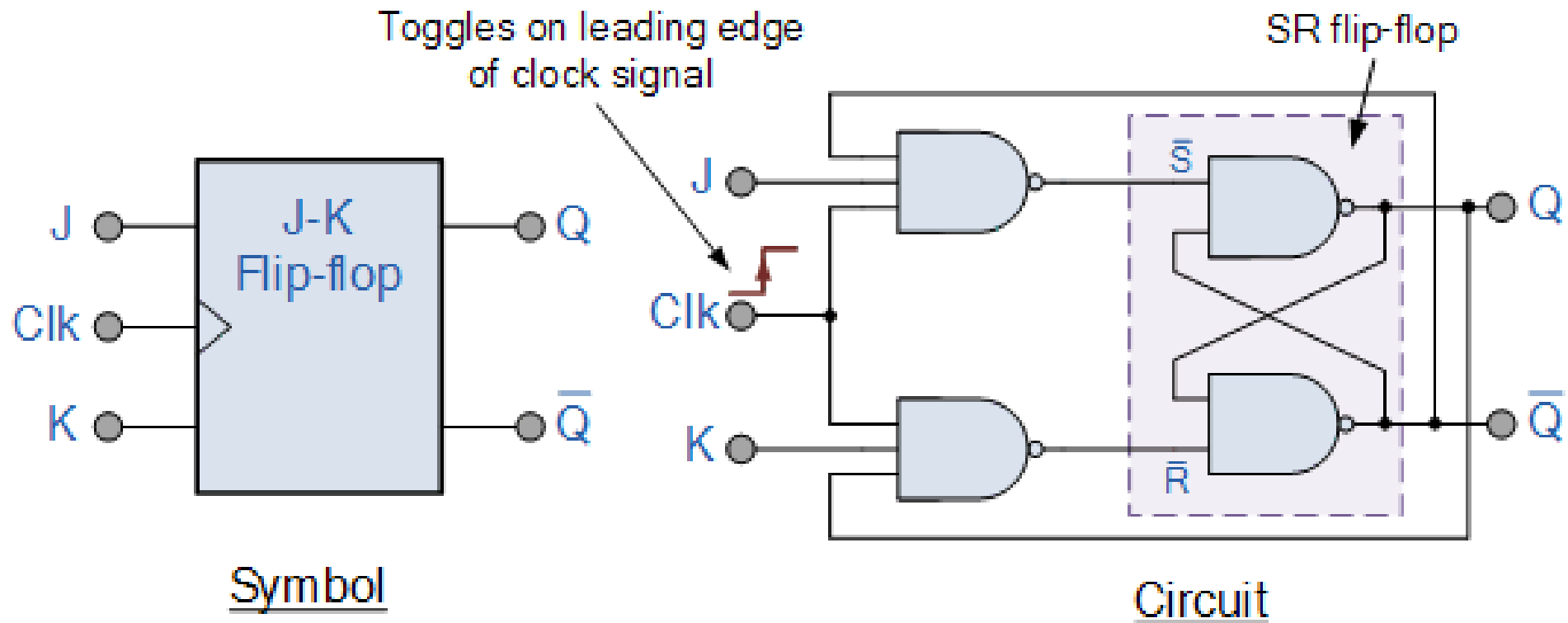
ใส่ Preset เป็น 0 (ทำงานเมื่อใส่ 0) เข้าไป $\rightarrow Q = 1$

NAND Clock RS Flip-Flop with Preset (PR) & Clear (CLR)

PR	CLR	CP	R	S	Q (t+1)	Q' (t+1)
0	0	X	X	X		
0	1	X	X	X		
1	0	X	X	X		
1	1	-	X	X		
1	1	↑	0	0		
1	1	↑	0	1		
1	1	↑	1	0		
1	1	↑	1	1		

JK Flip-Flop

A JK flip-flop is a refinement of the RS flip-flop in that the undefined state of the RS type is defined in the JK type.



JK Flip-Flop

J	K	Q (t)	Q (t+1)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Characteristic Table



Characteristic Equation

JK Flip-Flop

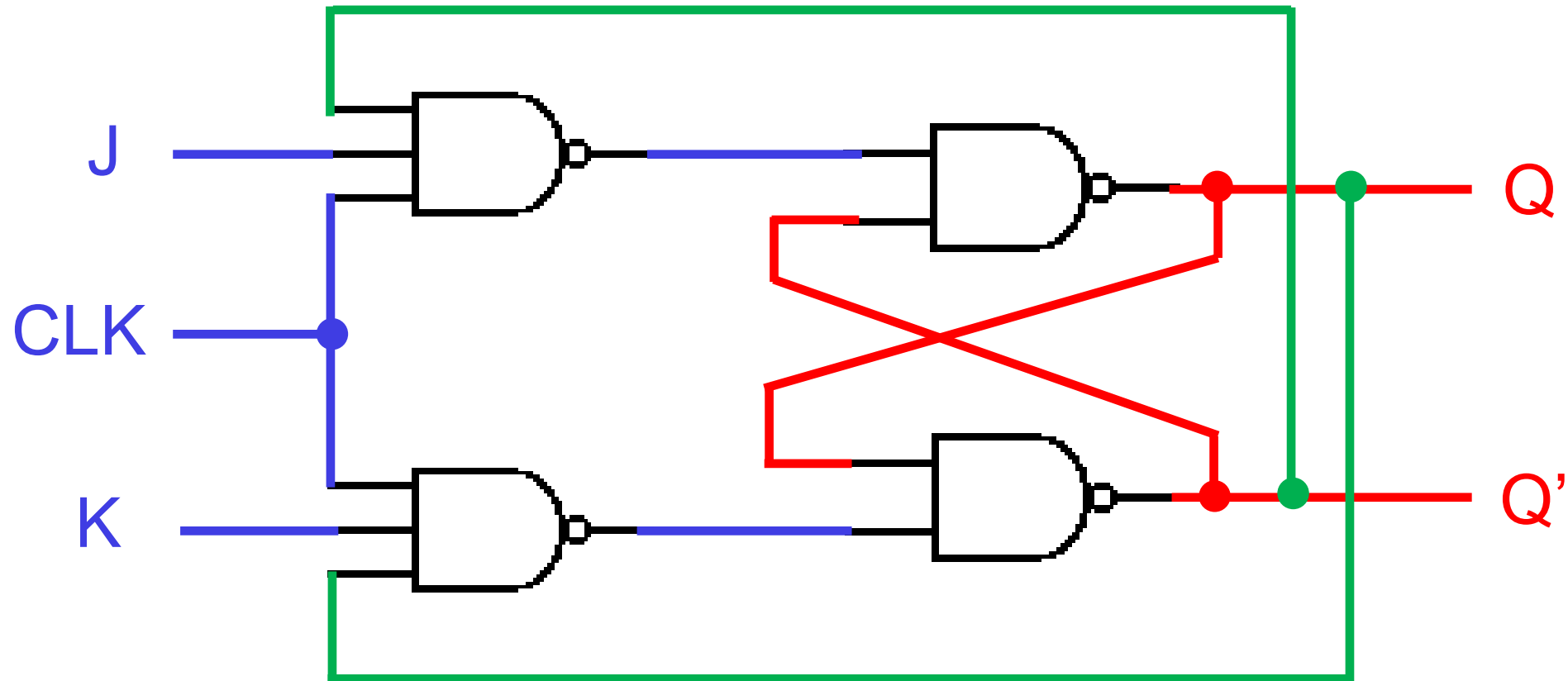
Q (t)	Q (t+1)	J	K
0	0		
0	1		
1	0		
1	1		

Excitation Table
(ตารางการกระตุ้น)

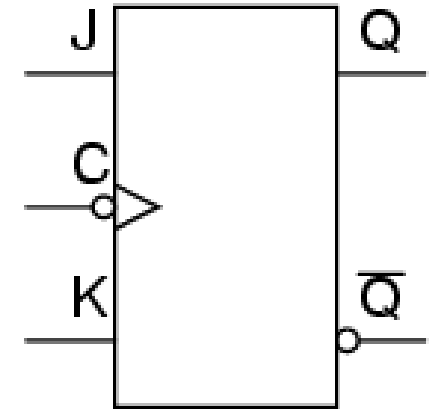
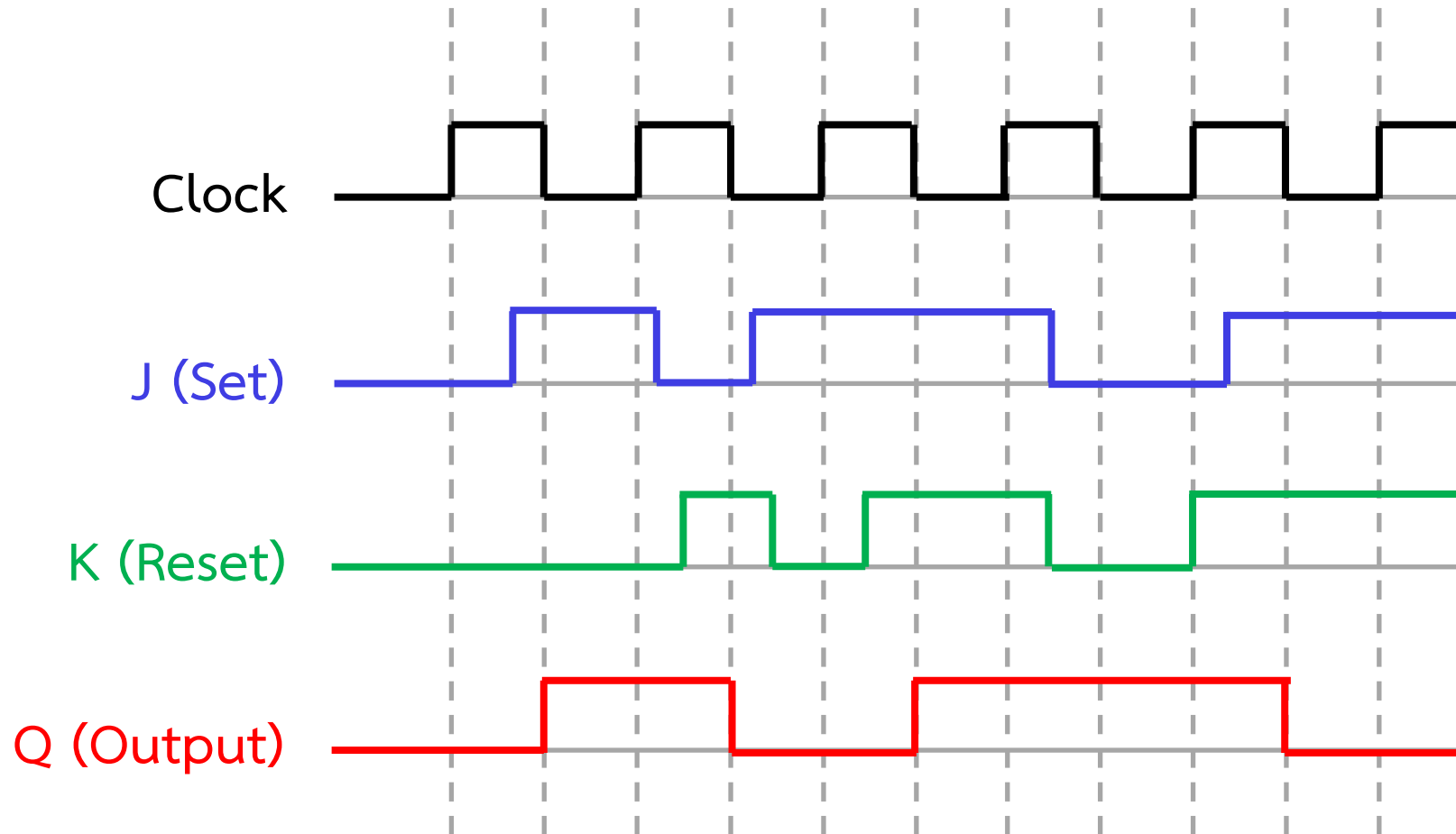
J	K	Q (t+1)	Q' (t+1)
0	0		
0	1		
1	0		
1	1		

Operation Table
(ตารางการทำงาน)

JK Flip-Flop



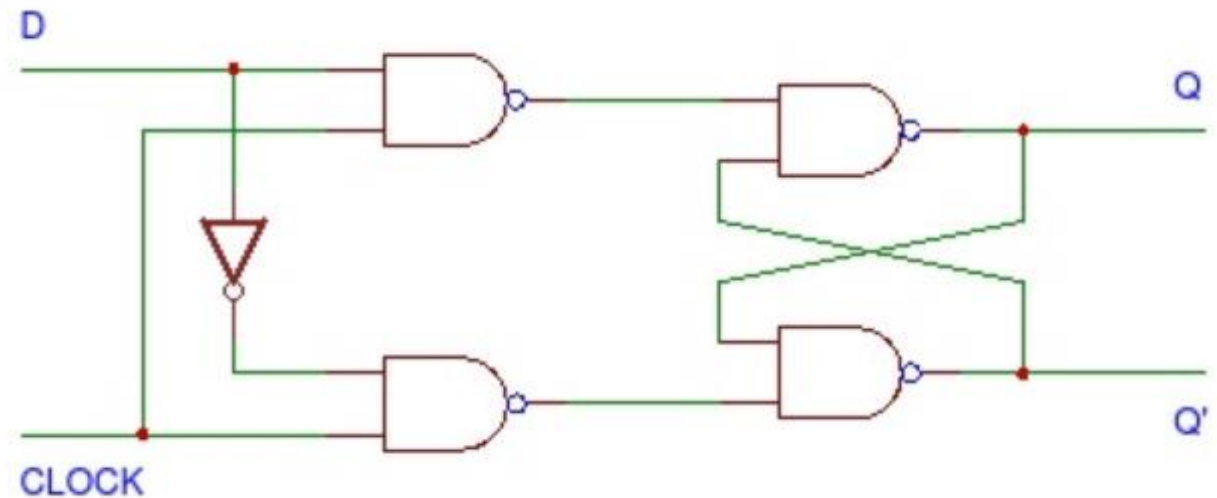
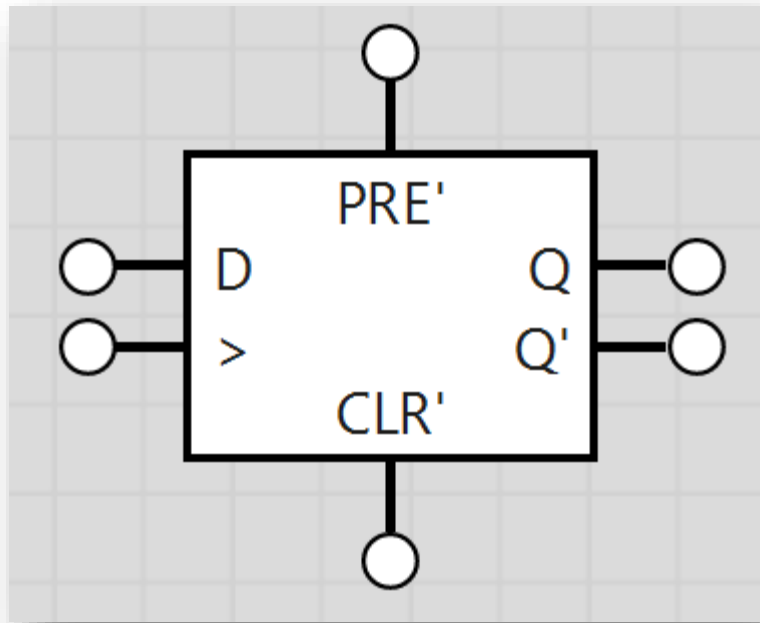
JK Flip-Flop Timing Diagram



เขียน Timing Diagram
ของ JK Flip-Flop ตัวนี้

D Flip-Flop

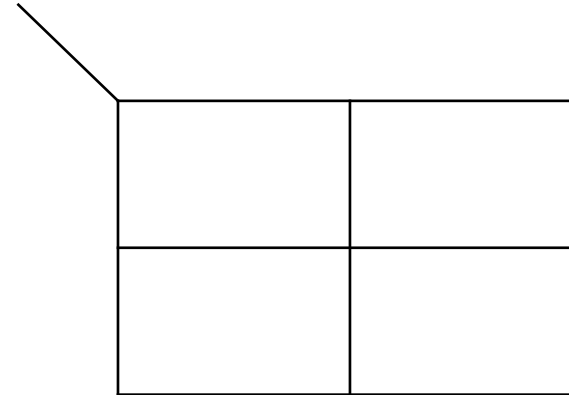
The D flip-flop is a **clocked flip-flop** with a single digital input 'D'. Each time a D flip-flop is clocked, its output follows the state of 'D'.



D Flip-Flop

D	Q (t)	Q (t+1)
0	0	
0	1	
1	0	
1	1	

Characteristic Table



Characteristic Equation

D Flip-Flop

Q (t)	Q (t+1)	D
0	0	
0	1	
1	0	
1	1	

Excitation Table
(ตารางการกระตุ้น)

D	Q (t+1)	Q' (t+1)
0		
1		

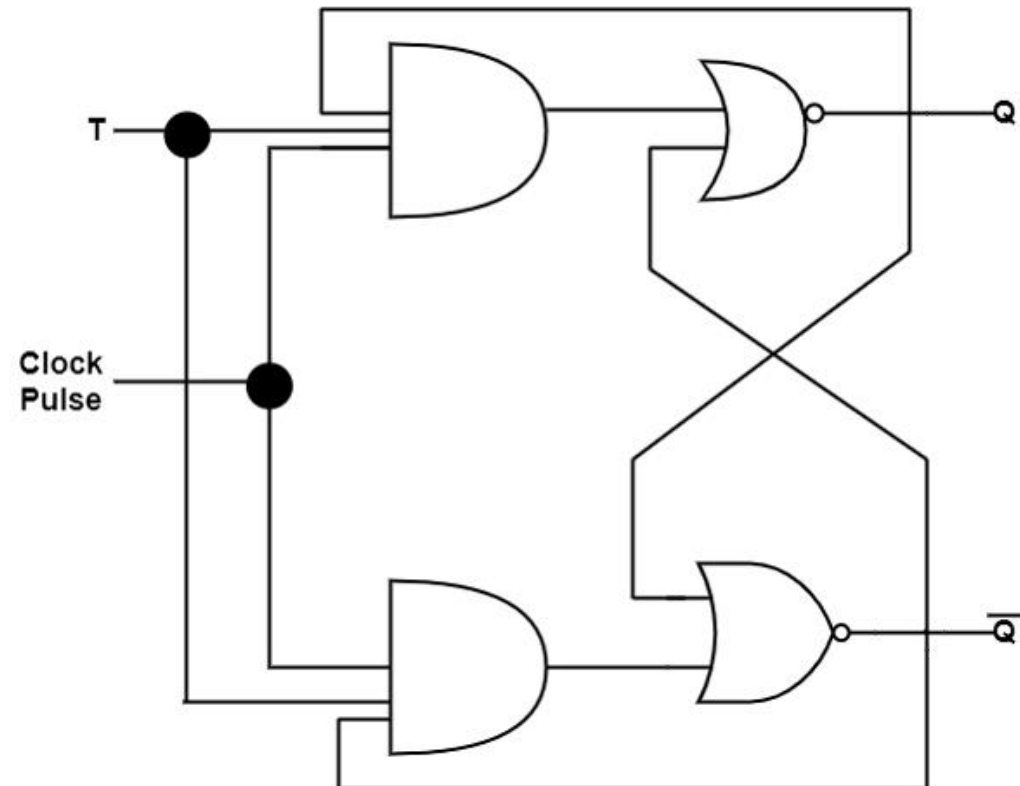
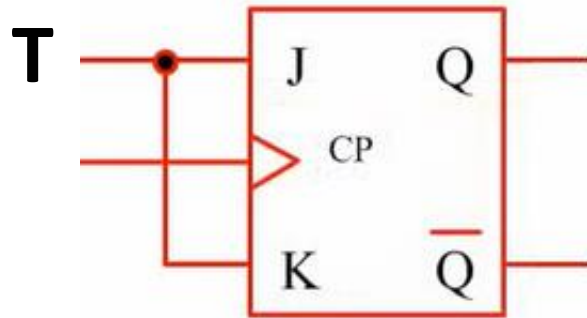
Operation Table
(ตารางการทำงาน)

T Flip-Flop

The T flip-flop is also called **toggle flip-flop**. It is a change of the JK flip-flop.

When $T = 0$, both AND gates are disabled. Therefore, there is no change in the output

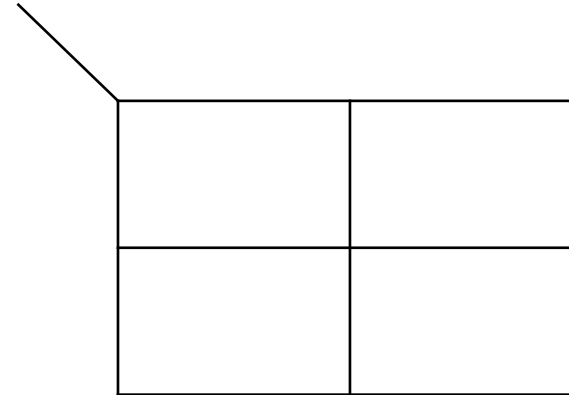
When $T = 1$, the output toggles



T Flip-Flop

T	Q (t)	Q (t+1)
0	0	
0	1	
1	0	
1	1	

Characteristic Table



Characteristic Equation

T Flip-Flop

Q (t)	Q (t+1)	T
0	0	
0	1	
1	0	
1	1	

Excitation Table
(ตารางการกระตุ้น)

T	Q (t+1)	Q' (t+1)
0		
1		

Operation Table
(ตารางการทำงาน)

Study without desire spoils
the memory, and it retains
nothing that it takes in.

Leonardo da Vinci

“ quote fancy

